Paleomagnetic Study of Igneous Rocks from the Manicouagan Structure, Quebec

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The Lower Triassic Manicouagan group is comprised of breccia, basalt diorite, and andesite, forming flat sheets partly filling a structural depression in Precambrian rocks. Cored, oriented samples show remarkably stable thermoremanent magnetization. Statistical analysis of the data shows that formation of the Manicouagan group has taken 10,000 years or less. Comparison with other Triassic pole positions from North America suggests that the area north of the St. Lawrence River has not rotated with respect to the rest of the continent since Triassic time.

Introduction. The octagon bounded by Manicouagan and Mushalagan lakes forms a conspicuous feature 59 km across incised into the rugged Precambrian terrane north of the St. Lawrence River. Ordovician sedimentary rocks overlain by flat-lying, unmetamorphosed, undeformed igneous rocks were discovered in this depression about 1944 [*Hammond,* 1945]. A whole rock K/Ar age determination on basalt from this structure gives an age of 225 million years [*Wanless et al.,* 1966]. This isolated occurrence of Lower Triassic rocks presents an unusually favorable opportunity for paleomagnetic studies to (1) determine any major post-Triassic tectonic movements in this region, (2) determine the approximate duration of igneous activity, (3) determine the pole position for the Lower Triassic period, assuming that the period of igneous activity was of the order of 10,000 years and that major tectonic movements have not affected the igneous rocks.

Geological setting. Figure 1 gives a geological sketch of the structure and the sampled localities. A detailed description of the geology is in press [*Currie and Murtaugh,* 1967]. Precambrian rocks in this region consist of granulite facies metamorphic rocks, with remnants of lower grade formations, cut by vast bodies of anorthosite and anorthositic gabbro. An octagonal block 59 km across has been down-dropped at least 1200 meters along systems of faults beneath the major lakes. These faults are of regional extent, some probably reaching as far as the St. Lawrence River. The inner edge of the warped, subsided mass is marked by fragments of disturbed Ordovician carbonate rocks up to 2 km in diameter occurring near the major lakes. The Manicouagan group of igneous rocks, comprising seven mappable units and at least five (probably many more) distinct cooling units, forms an annular plateau over 45 km in outside diameter and locally more than 250 meters thick, which rests on the down-dropped Precambrian. The central core of the structure consists of an anorthosite massif intruded by a sill. This central mountain is believed to be a horst faulted up at a late stage in the development of the structure.

The lowest parts of the Manicouagan group are comprised of highly altered, recrystallized, and deformed basement rocks, overlain by lenses and discontinuous sheets of glass-rich breccia almost identical to the suevite described from the Ries Basin, Germany [*Wagner,* 1965]. These rocks comprise less than 10% of the Manicouagan group but are of great interest because of unique petrographic features associated with them, including so-called 'shock metamorphism.' They were not sampled because of their heterogeneity.

Intruding the suevite, and forming sheets of limited extent is an aphanitic potash-rich black basalt. In two localities this material emanates from pipes a few hundred meters in diameter cutting vertically across the Precambrian. The rock consists principally of very fine-grained antigorite, with subordinate plagioclase microlites. Magnetite is present as minute, randomly distributed dust. The highly homogeneous character of this rock suggests that the present

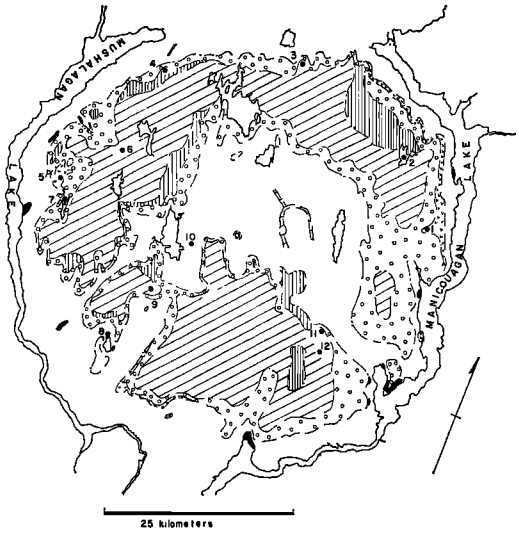


Fig. 1. Geological sketch of the Manicouagan structure showing sample sites. Checks, basalt, picrite; diagonal hatching, diorite; vertical hatching, medium-grained andesite; circles, fine-grained andesite; blank, Precambrian; solid black, Paleozoic.

mineralogy represents broad-spread deuteric alteration. This formation was sampled at localities 1 and 5 of Figure 1. The specimen whose age was determined was collected from a pipe of this material on the west shore of Manicouagan Lake.

Overlying and presumably younger than the basalt is a wide-spread subhorizontal sheet of coarse diorite. This material forms a sill more than 30 meters thick, underlying the anorthosite mass at the center of the structure. Occasional screens and remnants of basement rocks found on the igneous plateau suggest that in this region the diorite forms a complex of denuded sills. Spectacular porphyritic pyroxenes are the easy and characteristic identifying feature of this rock. Magnetite is present as large (1 mm) euhedral grains, with a rim of hematite forming about 10% of the grain. The hematite is so evenly developed that it is believed to form as a cooling phenomenon. Some regional variations in the degree of oxidation are mappable from the color of the rock and may represent different cooling units. These coarse grained rocks are well exposed and easy to drill. They were sampled at sites 2, 3, 6, 9, 11, and 12.

Intruding the diorite is a sequence of andesites forming a nearly flat-lying sheet underlying the diorite. Pigeonite is the most common pyroxene. The primary magnetite, distributed as fine dust, is partly converted to hematite along grain boundaries. As in the case of the diorite, this alteration is so universal and so evenly developed that it is interpreted to be a cooling phenomenon. There are several recognizable sub-units in the andesite, and two of them were sampled. Site 7 is a fine-grained buff rock with rare basement inclusions. Site 4 is a medium-grained black rock with little hematite. Both occur less than 6 meters above the basement. It was thought worth while to analyze the results separately to see if differences in cooling history might become apparent.

The Precambrian rocks within the structure are cut by numerous dykes of igneous rocks. In most cases they can be assigned to the diorites or andesites, but one group displays a glassy or cryptocrystalline mesostasis, is highly vesicular, and is characterized by strong metamorphic effects on the enveloping rocks. Magnetite (?) is present as exceedingly minute dispersed dust. This material was sampled at sites 8 and 10.

From time to time the Manicouagan structure has been alleged to be an impact scar [*Dence,* 1965] and the sampled rocks to be impact melt. The intrusive character of the igneous rocks, as well as other evidence [*Currie,* 1965], argues strongly against this interpretation, but for the present purpose it is unnecessary to decide on the origin of the structure. The petrographic and chemical homogeneity of the igneous rocks show that they have been totally melted and homogenized. Intrusive igneous activity persisted long enough so that the diorite was emplaced and completely crystallized to a grain size of several millimeters prior to intrusion of the andesites. Considering the thickness of the sill, it seems probable that this required several hundreds or thousands of years.

Sampling and magnetization measurements. An attempt was made to sample all units of the Manicouagan group sufficiently homogeneous to yield meaningful results, and to distribute the sample sites to yield reasonably uniform coverage of the structure. Because of difficult terrane it proved impracticable to

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collect samples in the southern quadrant of the structure. With this exception, the sampling objectives were obtained.

Fifty-one cores 2.54 cm in diameter were drilled from 12 sites (Figure 1). At each site, cores were drilled about 3 meters apart and oriented in situ with a solar compass. In most cases the cores were long enough to provide two cylinders 2.25 cm long. One core (from site 9) was discarded because of uncertainty about its orientation in situ.

The reported magnetization measurements were performed on a 3-magnet automatic astatic magnetometer, which yields magnetization intensities with an accuracy of ±1% and magnetization directions repeatable within 1°.

Magnetic stability tests. One specimen from each of the 12 sites was submitted to a series of alternating-field 'cleaning' tests on the apparatus of *Larochelle and Black* [1965]. The peak value of the demagnetizing alternating field was increased by successive steps of 50 oe from 100 to 550 oe and finally to 700 oe. The intensity of the natural remanent magnetization (NRM) and the magnetization vector determined after each treatment are plotted in Figures 2 and 3. The specimens displayed remarkable magnetic stability, except the specimen from site 11. The magnetic vectors (Figure 3) did not change significantly after the first treatment, although the vector of a few specimens (e.g. site 8) began to wander noticeably after treatment at 500 oe.

The specimen representative of site 11 was exceptional in three respects: (1) its NRM pointed steeply downward and east, (2) the intensity of the NRM was 40 times greater

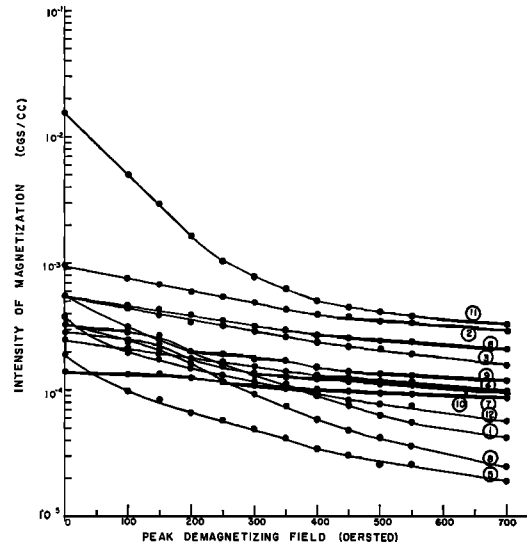


Fig. 2. Demagnetizing curves of twelve representative specimens. Circled numbers refer to site from which specimens came.

than the average of the other specimens and decreased abruptly after the first few cleaning treatments, (3) as the intensity of the peak demagnetizing field was increased, the direction of the magnetization vector crept slowly northward, i.e. toward the position of the magnetic vectors of the other specimens. After a final treatment at a peak field of 800 oe the declination of the vector was 61.4° and its inclination 51.6°. These results indicate the presence of a strong component of isothermal remanent magnetization in the specimen, probably due to a lightning strike in the geological past.

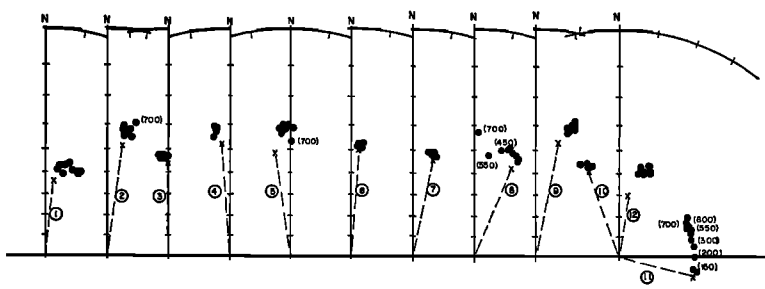


Fig. 3. Stereographic projections of magnetization measured after individual treatment. Crosses are the NRM directions; circled numbers refer to site of individual specimen; numbers in parentheses are peak field amplitude of treatments.

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Natural and stable remanent magnetization. The intensity and direction of the NRM was determined for 50 cores. The direction of magnetization of a core was obtained by averaging results from the two 2.25-cm-long cylinders cut from it, except for cores from site 8 (Table 1, column 3) which yielded only one cylinder. The angle θ between the two vectors rarely exceeded 5° and, with the exception of cores from site 11, never exceeded 10°. These results establish the magnetic homogeneity of the rocks.

After measurement of the NRM, all cylinders were submitted to two magnetic field treatments in peak demagnetizing fields of 100 and 200 oe, respectively. Magnetization measurements after the second treatment showed that the unstable component of viscous magnetization originally present in some of the cores had been completely removed by the first treatment. The data are summarized in Table 1, where the mean magnetic vectors for each site and their corresponding angular variance estimates $\hat{\delta}^2$ [*Larochelle*, 1967] are listed.

Table 1 shows that (1) the angles of inclination generally decreased after the first treatment, and the angles of declination generally increased, (2) the clustering of vectors at each site was not significantly altered by either cleaning treatment, (3) the mean direction of magnetization at site 11 followed the trend indicated by the test specimen (Figure 2). After cleaning at 100 and 200 oe, and a third treatment in the peak demagnetizing field of 800 oe, the declination of this site had crept to 67.6°, whereas its inclination (50.2°) and its angular variance estimate (0.016) were not modified significantly. The unstable magnetizations observed for this site were considered unrelated to the ambient earth's field at the time the rocks were originally magnetized and are consequently not included in the following statistical analysis of the data.

Analysis of the stable remanent magnetization directions. The within-site and between-site angular variance estimates [*Larochelle*, 1967] were calculated for the directions of magnetization obtained after the 200-oe treatment. These parameters are defined respectively by

$$\hat{\delta}_{ws}^{\,2} = 2(N - \Sigma R_i)/(N - B)$$

and

$$\hat{\delta}_{bs}^{\,2} = 2(\Sigma R_i - R)/(B - 1)$$

where R and R_i are respectively the moduli of the resultants of all N vectors in the sample and

TABLE 1. Magnetization Directions

				After Alternating-Field Treatment at								
				NRM			100 oe			200 oe			
Unit	Site	n	N	D	I	$\hat{\delta}^2$, radians^2	D	I	$\hat{\delta}^2$, radians^2	D	I	$\hat{\delta}^2$, radians^2	θ_{av}
1	1	···	5	14.7	57.7	.00901	17.5	47.6	.00913	19.2	46.7	.00722	5.2
2	2	···	3	12.2	46.6	.05263	11.9	39.9	.03226	11.6	41.1	.03125	4.7
2	3	···	6	353.3	47.6	.02941	353.9	45.9	.03007	353.6	45.7	.03509	2.2
3	4	···	2	2.3	38.0	.01818	0.5	34.7	.02299	359.7	33.2	.01460	1.6
1	5	···	4	17.4	49.9	.07407	19.7	40.4	.06061	20.0	39.6	.07407	4.7
2	6	1	6	11.1	40.7	.01299	11.0	39.5	.01361	11.3	39.7	.01351	1.3
4	7	···	6	15.1	39.9	.00266	15.4	38.7	.00209	15.6	40.0	.00242	3.8
5	8	1	1	23.5	44.4	···	25.6	40.9	···	24.3	39.8	···	···
2	9	2	3	12.2	40.1	.01639	14.8	35.3	.01399	16.6	34.3	.01333	2.1
5	10	2	2	358.8	45.9	.09524	0.4	44.8	.11765	0.9	44.4	.08696	···
2	11	2	6	95.4	47.9	.02174	95.0	49.6	.01942	90.5	50.7	.01351	11.1
2	12	···	6	16.2	62.3	.00497	18.8	44.8	.00980	18.8	44.3	.00885	4.0

Explanation: N is the number of cores; n is the number of cores represented by only one cylinder; D and I are the declination and inclination of the site mean direction of magnetization; $\hat{\delta}^2$ is the angular variance estimate [*Larochelle*, 1967]; θ_{av} is the average angle between the magnetic vectors for the two cylinders cut from individual cores.

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TABLE 2. Within- and Between-Site (Unit) Variance Estimates

i	B_i	N	$\hat{\delta}_{wi}^2$, radians^2	$\hat{\delta}_{bi}^2$, radians^2	$\hat{\delta}_{bi}^2/\hat{\delta}_{wi}^2$	$F_{2(B-1),\ 2(N-B),\ 0.05}$
s	11	44	0.01988	0.08097	4.08	1.75
u	5	44	0.03311	0.06514	1.93	2.07

Explanation: $\hat{\delta}_{ws}^2(\hat{\delta}_{wu}^2)$ within-site (unit) angular variance estimate; $\hat{\delta}_{bs}^2(\hat{\delta}_{bu}^2)$ between-site (-unit) angular variance estimate; B_i number of sites (units).

TABLE 3. Mean Unit Directions of Magnetization and Corresponding Virtual Poles

Sites	Unit	Magnetizations (200 oe)				Pole Positions			
		N	D	I	$\hat{\delta}^2$, radians^2	Longitude, °E	Latitude, °N	$\delta\chi$,* deg	$\delta\psi$,* deg
1, 5	1	9	19.6	43.6	0.03571	73.6	60.1	8.7	5.4
2, 3, 6, 9, 12	2	24	09.8	42.2	0.03571	92.2	62.1	4.9	3.0
4	3	2	359.7	33.2	0.01460	111.5	56.6	24.4	13.9
7	4	6	15.6	38.0	0.00242	83.2	57.6	2.8	1.6
8, 10	5	3	9.2	43.4	0.07692	92.7	63.0	30.6	19.0
Mean of 44 cores		44	12.0	41.7	0.03636	88.2	61.1	3.6	2.2
Mean of 5 units		5	10.5	40.3	0.01639	91.4	60.4	8.4	5.1
Mean Pole		5	...	...	0.01802	90.9	60.5	...	...

* $\delta\chi$ and $\delta\psi$ are the axes of the oval of confidence about the poles at the 95% significance level.

of the N_i vectors representing the ith of the B sites. The corresponding statistics calculated with the geological unit as the subgroup will be represented as $\hat{\delta}_{wu}^2$ and $\hat{\delta}_{bu}^2$, respectively.

It is shown [*Larochelle*, 1967] that, if

$$\hat{\delta}_{bs}^2/\hat{\delta}_{ws}^2 \leq F_{2(B-1),2(N-B),\alpha}$$

the probability that the site mean directions are significantly different is equal to or less than α. The quantity to the right of the inequality sign refers to the statistic F listed in mathematical tables for $2(B-1)$ and $2(N-B)$ degrees of freedom at the probability level α. As shown in Table 2, the site mean directions are significantly distinct at the 0.05 probability level, whereas the unit mean directions are not. Slight differential movements from site to site within units could explain these results.

In Table 3, the 44 directions of magnetization are grouped on the basis of the geological unit they represent, together with the corresponding angular variance estimates and the virtual geomagnetic pole positions derived from each unit. The center of gravity and the angular variance estimates of the five virtual poles are also listed on the last line of the table.

As the mean directions of magnetization of the geological units are not significantly different, the direction of their mean was used in

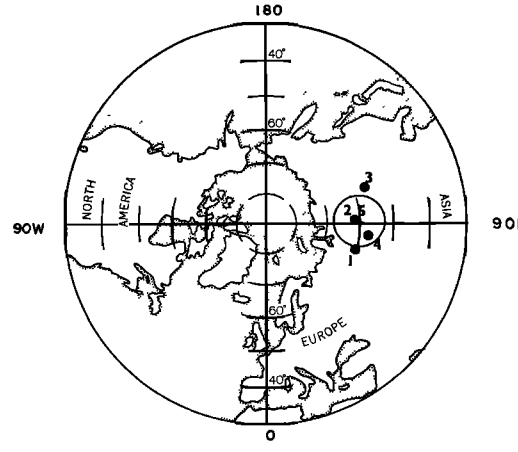


Fig. 4. Stereographic projection of five virtual poles. Full circle centered on their mean is their circle of confidence of radius α_{95}.

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TABLE 4. Pole Positions Derived for Various North American Triassic Formations

Formation	Pole Position		N*	$\delta\chi$, deg	$\delta\psi$, deg	Reference
	Longitude, °E	Latitude, °N				
Nova Scotia lavas	75½	77	12	11½	11½	Bowker in *Cox and Doell* [1960]
Newark series	108	63	29	4	3	*Opdyke* [1961]
Connecticut rocks	86	54	—	—	—	*DuBois et al.* [1957]
Massachusetts lavas	90	55	—	—	—	*Irving and Banks* [1961]
Moenkopi	103	62	—	—	—	*Collinson and Runcorn* [1960]
Chugwater	112½	48	—	—	—	*Collinson and Runcorn* [1960]
Pennsylvania dikes	105	62	20	3	2	*Beck* [1965]
Nova Scotia dike	98	69	2	4	2½	*Larochelle and Wanless* [1966]
Manicouagan group	88½	60½	11	6	3½	This paper

* N is the number of site means used to calculate the pole positions.

calculating the virtual pole position during the cooling period of the Manicouagan group. The units means have an angular standard deviation of 7.3°, which is considerably smaller than the 15° standard deviation that would be expected if the units had been magnetized intermittently over a period of at least 10,000 years at magnetic latitude of 23.5° [*Cox*, 1962; *Creer*, 1962; *Gough et al.*, 1964]. This confirms the view that the five units of the Manicouagan group were probably ejected within a period considerably shorter than 10,000 years. It is interesting to note, however, that, if the virtual poles obtained for each unit are plotted on a polar projection (Figure 4), they appear to define a nearly closed loop about 2100 km by 500 km. If this is due to secular variation of the magnetic pole, unit 5 is of the same age as unit 2, and units 3 and 4 are of perceptibly different ages. These requirements are in accord with the field data, although the coincidence of poles of units 2 and 5 may be fortuitous, owing to the rather dispersed data used for determination of the pole of unit 5.

In Table 4 the mean pole position of the Manicouagan igneous rocks (88.5°E, 60.5°N) is compared with pole positions derived from some other North American Triassic formations. The slight discrepancy between this Lower Triassic pole and the Upper Triassic pole positions is probably not significant. Neither the hypothesis of gradual polar wandering nor the rotation of Quebec with respect to the rest of North America is supported by these data.

Summary and conclusions. The petrographic and magnetic homogeneity of the hematite-bearing cores suggests that this constituent is deuteric and that the stable remanent magnetization is of thermoremanent origin.

The emplacement and cooling of the Manicouagan group probably required less than 10,000 years. The late andesitic rocks (units 3 and 4) comprise two or more cooling units.

The mean pole position for the Manicouagan group is in good agreement with the mean pole positions derived from other North American Triassic formations.

This coincidence suggests that the region north of the St. Lawrence River has not suffered any major horizontal movement with respect to the rest of North America since the early Triassic period.

Between-site dispersion in magnetization directions confirms the geological deduction of small-scale vertical adjustments between blocks within the Manicouagan structure.

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